

Embracing software-defined innovation in technology projects

At the heart of the Bobby Car initiative was the desire to transition from a hardware-centric worldview to one that is software-defined. This shift required reimagining everything from the operating system to the deployment of containers and microservices in various applications. The project demonstrated how to accelerate deployment cycles by leveraging cloud-native technology principles, thus bringing a new level of agility and adaptability to traditionally rigid industries.

One of the significant challenges addressed by Bobby Car was establishing a robust base that adhered to functional safety standards, beginning with the operating system. Extending this approach through the entire value chain, the project facilitated the development and testing of new features, ensuring seamless integration without disrupting existing systems. This methodology was crucial, especially for connected vehicle platforms and IoT architectures, where reliability and safety are paramount.

A key aspect of the Bobby Car project was its emphasis on learning from past experiences and best practices. This knowledge was crucial in developing a strategy that was not only innovative but also practical and sustainable. Continuous integration and deployment became fundamental, as was the ability to conduct virtual testing and create digital twins to enhance efficiency and provide improved solutions to customers.

A pivotal element of the Bobby Car initiative was also its integration with a broader ecosystem, encompassing everything from payment systems to 5G technology and the manufacturing sector. This comprehensive approach ensured that applications developed under the Bobby Car project could seamlessly interact with various components of the industry,

enhancing the user experience and operational efficiency.

Key components and applications

Bobby Car consists of two primary components: the “vehicles waiter” and the “geography console.” The vehicles waiter is an intricate system providing detailed insights into various metrics such as delivery data, temperature, speed, RPM, and more, leveraging data from the OBD port of vehicles. This feature is crucial for real-time data analysis and decision-making.

The geography console, on the other hand, is implemented as a series of Kubernetes clusters. This setup enables the deployment of CRD objects, facilitating the ability to introduce isolated objects within the software stack. This aspect is particularly beneficial for conducting over-the-air (OTA) updates, allowing for modifications without disrupting other parts of the system.

One of the most significant advantages of Bobby Car is its data handling capability. The project utilises a DevSecOps pipeline, ensuring that data application and deployment are both efficient and secure. This pipeline is adaptable to the existing systems within an organisation, making Bobby Car a versatile tool for various business needs.

A notable application of Bobby Car is its air interface, equipped with a feature known as ‘Dark Spot.’ This involves a unique interaction with unit drivers, adding another layer of complexity and utility to the project. Furthermore, the geography console of Bobby Car not only handles deployment zones but also incorporates location-based services, utilising latitude and longitude for precise coordination.

Bridging software and hardware in connected cars

Bobby Car stands out for its ability to seamlessly interact with IoT devices and hardware. This project, completed in

collaboration with various organisations, including BMW and ECS, has pioneered a pattern for creating connected cars. The essence of Bobby Car is in its integration of software with the physical components of a car, embodying the intersection of digital and mechanical engineering.

The Bobby Car project is centred around three essential elements. First, it incorporates sensors that act as the project’s eyes and ears, gathering environmental data. Next is the software interaction aspect, where Bobby Car’s software is specifically designed to efficiently interact with these sensors, ensuring effective data processing and utilisation. Finally, the project is grounded in its robust infrastructure, which forms the backbone by supporting and running both the software and the sensors seamlessly.

Key technologies powering Bobby Car

Bobby Car utilises several core technologies to enhance its functionality:

OpenShift platform: The platform is pivotal to the project, allowing the software to efficiently scale from managing a few IoT devices or cars to potentially thousands or millions. It guarantees uniform performance across diverse deployment environments, including on-premises and cloud settings, thereby ensuring a smooth transition across these platforms. This consistency and scalability are fundamental to the project’s success across different scenarios.

Motor vehicle simulator and MQTT protocol: These are used for real-time data transmission. The MQTT protocol, in particular, is crucial for its reliability even in unstable network conditions, making it ideal for IoT devices that require consistent connectivity.

BAFTA messaging protocol: This complements MQTT by providing a robust messaging infrastructure capable of handling large volumes of data from thousands of IoT devices.